BOARD DECISION MEMO / FY2025 CAPITAL ALLOCATION

Capital allocation review

Emerging-market expansion vs. disruptive technology R&D

TO	Board of Directors	**FROM**	Senior Finance Manager
DATE	January 2025	**SUBJECT**	FY2025 investment decision

Executive recommendation

Advance Project B as the stand-alone first choice - with stage gates.

The illustrative, probability-weighted base case indicates approximately $33 million of NPV at the 9% WACC and about 22% IRR on expected cash flows, versus roughly $20 million NPV and 16% IRR for Project A. Project B's 30% failure risk is material; authorize a capped, milestone-based program, not an unconditional $40 million release. The underlying brief does not provide revenue bases, margins, taxes, or full cash-flow forecasts, so these are directional estimates, not a valuation.

PROJECT A | BASE CASE	PROJECT B | EXPECTED CASE	PORTFOLIO | BOTH
~$20m NPV | ~16% IRR	~$33m NPV | ~22% IRR	~$53m combined NPV*

What the Board is being asked to decide

- If choosing one venture: authorize Project B to proceed through a gated development plan, subject to technical, customer-validation, and cost-to-complete checkpoints.
- If pursuing both: set a $50m cap for Project A and $40m cap for Project B; retain the remaining $10m as a centrally controlled liquidity / contingency reserve.
- Require a refreshed cash forecast and full project-level cash-flow cases before final funding release. The $100m cash figure is treated here as the planning envelope despite the prompt's May 2025 date; Treasury should validate cash actually available at the January 2025 decision date.

**Combined central-case NPV is a simple sum, assumes no interaction/cannibalization and is not adjusted for portfolio risk. See the modeling basis and limitations on page 2.*

01 / FINANCIAL VIEW

Directional NPV and IRR assessment

The brief lacks forecast revenue dollars, margins, launch timing, taxes, working capital, and maintenance capex, so exact NPV / IRR cannot be derived. Finance uses a 10-year unlevered free-cash-flow proxy at the 9% WACC with no terminal value. Estimates are directional and not a substitute for a full investment case.

Model input	Project A	Project B	Interpretation
Initial investment	$50m	$40m	As stated in the brief; treated as time-zero outflow.
Proxy Year 1 FCF	$6m	$6m if successful	Illustrative cash contribution proxy, not provided revenue.
Growth profile	20% annually in Years 2-5; 5% in Years 6-10	50% in Years 2-3; 20% in Years 4-5; 5% in Years 6-10	Applies described revenue growth to FCF as a simplified proxy.
Risk treatment	Execution / market risk reflected in sensitivity range	70% success / 30% failure; failure assumes no material operating cash flow	No salvage or recovery value assumed in failure case.
Discount / horizon	9% WACC / 10 years	9% WACC / 10 years	No terminal value; no strategic spillover value.

Indicative outputs

Project / case	NPV at 9%	IRR direction	Main interpretation
A - central proxy	~+$20m	~16%	Above WACC, but value depends on executing the launch and converting growth into cash.
A - proxy range	~+$8m to +$43m	~12% to 23%	Based on Year 1 FCF of $5m-$8m; weak launch economics or delay can erase value.
B - expected case	~+$33m	~22% expected-cash-flow IRR	70% probability-weighted cash flows; not the statistical expected IRR.
B - successful case	~+$64m	~31%	If the $6m Year 1 success-case FCF proxy and growth ramp occur.
B - failure case	~-$40m before recovery	Not meaningful	If the full investment is lost and no material cash flows result.
B - expected range	~+$10m to +$80m	~13% to 36% expected-cash-flow IRR	Year 1 success-case FCF proxy of $4m-$10m; wide outcome dispersion.

Decision signal

Both central proxy cases exceed the 9% hurdle. B has higher expected value but a wider distribution: 30% failure could erase nearly $40m. Break-even success probability is about 38% under the modeled success cash flows, versus the stated 70%.

Sensitivity: at 12% / 15% WACC, central proxy NPV is about $10m / $2m for A and $22m / $14m for B. Results remain sensitive to cash conversion, delay, and technical success. Rounded; directional only.

02 / STRATEGIC CHOICE

Why Project B is the preferred first commitment

On a stand-alone basis, Project B is the better first commitment under the stated assumptions. The conclusion is not that B is safer; it is that the company's existing technical capability and the project's strategic upside can justify a bounded, actively managed risk.

Dimension	Project A - emerging markets	Project B - disruptive R&D
Financial profile	Moderate-growth opportunity; proxy NPV is positive but lower. Growth converts to value only if local launch economics and infrastructure utilization hold.	Higher expected NPV and upside if successful; 30% failure probability creates meaningful capital-loss exposure.
Strategic alignment	Strong diversification into new geographies, customers, and product line; reduces concentration in the existing market over time.	Defends / extends leadership in the core market; leverages strong internal R&D team and could create a differentiated platform or new category.
Execution fit	Requires new infrastructure and local partnerships; limited company experience in target markets. More exposed to geopolitical, regulatory, and FX risks.	Builds on existing expertise and relationships; more directly within company capabilities. Exposed to technical, IP, and incumbent-response risks.
Long-term value	Adds geographic breadth and potential recurring customer relationships; may create options for adjacent products.	Potentially creates defensible IP, pricing power, platform effects, and a step-change in market position; upside may be nonlinear.
Assessment	Attractive diversification option; better suited as a staged second leg once market-entry economics are validated.	Preferred first bet, provided spend is gated and management is willing to stop if evidence deteriorates.

Board rationale and guardrails

- Value creation: the base proxy suggests B's expected NPV exceeds A's by roughly $13m. B may also create strategic assets not captured in the NPV model, including IP, product differentiation, and future platform revenue.
- Capability fit: B uses the firm's established R&D base and core-market knowledge; A requires new local infrastructure and partnerships where experience is limited.
- Risk capacity: strong profitability, balance sheet, and stated $100m liquidity provide capacity to absorb a bounded loss, but do not make a 30% failure probability acceptable without controls.
- Portfolio balance: A's geographic diversification is valuable and helps counterbalance B's technology / core-market exposure. The recommendation is a sequencing choice, not a rejection of A.
- Capital discipline: release B funding against evidence of technical performance, customer willingness to pay, and a credible cost-to-complete. Failure at a gate should trigger stop / pivot rather than automatic follow-on funding.

Recommendation in one sentence

Approve Project B as the preferred stand-alone investment, subject to stage gates and a hard $40m cap; if the Board elects to pursue both viable ventures, fund both within a $90m project envelope and keep $10m uncommitted for liquidity and contingencies.

03 / PROJECT B RISK PLAN

Top three financial and operational risks

The following risks are ranked by potential to impair the investment thesis or consume capital. Mitigations should be embedded in the delivery plan, with the Board receiving a short quarterly dashboard.

Risk / exposure	Why it matters	Mitigation actions	Early-warning indicators	Contingency if triggered
1. R&D failure or technical obsolescence | High	A technical miss, slow performance, or a faster competing architecture could strand much of the $40m and leave little recovery value.	Break work into gated prototype, validation, and scale phases; set measurable performance thresholds and independent technical reviews; benchmark emerging alternatives; maintain a focused minimum viable product and stop-work criteria.	Missed technical milestones; reliability below target; repeated rework; falling performance / cost advantage versus alternatives; rising cost-to-complete.	Stop or narrow scope at the next gate; pivot to the most valuable adjacent use case; license / partner for missing technology; preserve reusable IP and redeploy uncommitted funds.
2. Weak adoption / competitive response	The >50% growth potential is conditional on success and adoption. Incumbents may cut prices, bundle substitutes, or lock up customers; weak demand reduces revenue and delays payback.	Secure design partners and paid pilots early; test willingness to pay and use cases before full-scale launch; plan differentiated positioning; build customer feedback into product milestones; prepare pricing and channel scenarios.	Pilot conversion and repeat use below plan; customer acquisition cost rising; sales-cycle extension; lost design wins; competitor price cuts or accelerated launches.	Narrow launch to a defensible niche; adjust packaging / price; pursue a distribution or co-development partner; delay scale spend; license the technology if direct go-to-market economics weaken.
3. IP, schedule, and cost-to-complete risk	IP leakage / infringement can erode advantage or create litigation; talent, supply, or integration delays can raise spend and push cash generation beyond the hurdle window.	Complete freedom-to-operate review and patent / trade-secret plan before disclosure; secure code and lab access; use milestone-based vendor contracts; maintain a named critical-path owner, contingency in schedule, and monthly estimate-at-completion review.	Unresolved patent claims; key-person attrition; missed integration dates; vendor change orders; forecast-to-complete rising >10%; slipping customer launch commitments.	Redesign around protected technology or license it; use alternate suppliers / contractors; reduce launch scope; pause hiring and noncritical spend; return for Board approval before using reserve or exceeding the $40m cap.

Control cadence

- Monthly: technical milestones, paid-pilot pipeline, spend vs. approved budget, estimate-at-completion, key-person and IP issues.
- At each funding gate: independent review of remaining technical risk, updated probability of success, probability-weighted NPV, and realistic alternatives including stop / partner / license.
- Quarterly to the Board: red/amber/green dashboard and explicit request for any scope, timing, or capital change. No automatic top-up if cost-to-complete exceeds the cap.

04 / IF BOTH PROJECTS PROCEED

Recommended allocation of the $100 million

If the Board concludes both ventures are viable and chooses to pursue both, fund the project envelopes in full but preserve a modest central reserve. The two stated project budgets total $90m, so there is no need to force the remaining $10m into projects before evidence supports it.

Use of funds	Allocation	Share of $100m	Capital treatment
Project A - emerging-market launch	$50m	50%	Approve as capped project envelope; release large infrastructure commitments only after country, partner, regulatory, and unit-economics diligence.
Project B - disruptive R&D	$40m	40%	Approve as capped project envelope; stage releases against technical and customer milestones.
Central reserve / liquidity	$10m	10%	Retain centrally for working-capital volatility, unexpected but value-preserving needs, or liquidity protection; not pre-authorized project overrun.

Portfolio value - and its limits

Under the central proxy cases, combined NPV is about +$53m ($20m A + $33m B). The projects broaden exposure across geographies and customer segments (A) and technology / product positioning (B), supporting long-term value creation. This is not a risk-free diversified portfolio: both remain exposed to technology demand and competitive cycles, and the combined NPV assumes no cannibalization or execution interaction. If B fails with no recovery, A's roughly +$20m central NPV would not fully offset B's -$40m loss.

How to preserve option value and liquidity

- Treat $50m / $40m as caps, not spend targets. Gate commitments; for A, complete country, partner, regulatory, and unit-economics diligence and secure exit / drawdown protections before infrastructure contracts.
- Keep the $10m reserve under CFO / Board control for liquidity or proven value-preserving contingencies, not to rescue a weakening business case.
- Maintain separate project P&Ls; refresh cash flows quarterly using actual revenue, margin, tax, working capital, capex, and launch timing.
- If B misses a gate, stop / pivot and retain unspent cash; move capital to A only after validation and incremental-return review.
- Both projects diversify execution drivers, but preserve operating liquidity and debt capacity; a strong balance sheet is not a reason to exhaust cash.

Final approval gate: Treasury confirms available cash and minimum liquidity; management provides source-based cash-flow cases. Before follow-on, the Board reviews customer / technical evidence, refreshed risk-adjusted NPV and cost-to-complete, and material FX, regulatory, IP, or competitor changes.

Source: Board-provided project brief; Finance proxy calculations and judgments. USD millions unless noted. Estimates rounded; no external market data used.